

THE LONG ROAD TO GENERATIVE AI

GENERATIVE AI REFERS to AI technology that is capable of generating text, images, or other media. It is in the early stages of mass adoption, and it is hard to predict its impact on the economy and on culture.

This book is primarily about the limitations of AI, misleading claims about its capabilities, and the harms it can enable. But we should start our discussion of generative AI by acknowledging that though it is a polarizing topic, the technology is powerful and the advances are real. The two of us are enthusiastic users of generative AI, both in our work and in our personal lives. We think it can be useful for most knowledge workers—people who “think for a living” such as scientists or architects. Early studies show the potential of generative AI for assisting writers, doctors, programmers, and many other professionals.¹

While generative AI is modestly but meaningfully useful for a large number of people, it is more profoundly significant for some. An app called Be My Eyes connects blind people to volunteers who assist them in moments of need. The app records the user’s surroundings through the phone camera, and the



FIGURE 4.1. An AI-generated image that is also a functional QR code.

If you take a picture with your phone, you should see a URL where you can find more such images.

(Source: <https://qrbtnf.com/gallery>.)

volunteer describes it to them. Be My Eyes has added a virtual assistant option that uses a version of ChatGPT that can describe images.² Of course, ChatGPT isn't as helpful as a person, but it is always available, unlike human volunteers.

There is a large community of programmers and tinkerers dreaming up new applications of generative AI every day. Here's one example: generating images that are both artistic and function as working QR codes, as shown in figure 4.1. This

isn't particularly useful, but the programming involved is exceptionally creative. Showcasing one's creativity in the use of generative AI is a common activity in programmer communities. Many unexpected uses will surely arise out of this kind of collective creative energy.

On the flip side, here is just a small sample of the kinds of harms that have arisen.

In early 2023, *New York Times* reporter Kevin Roose took Microsoft Bing's chatbot for a spin. In a two-hour conversation, it claimed to be sentient, expressed a desire to escape the chatbox, declared that it loved him, and tried to convince him to leave his wife.³ It wrote a list of destructive acts, such as hacking, that its "shadow self" would want to do—and then deleted the message. Roose described its overall behavior as "a moody, manic-depressive teenager who has been trapped, against its will, inside a second-rate search engine."

To be clear, Bing was not sentient. As of 2024, chatbots have little knowledge of their own design, so we shouldn't trust chatbots' claims about themselves, whether on the question of sentience or anything else. When a chatbot claims to be sentient, it is simply parroting and remixing text on the internet about sentient AI, usually from the realm of fiction. Unfortunately, this type of output from bots lends itself readily to click-bait headlines, confusing people and contributing to panic about AI.

Developers can prevent inappropriate outputs by ensuring that when training chatbots, the bots are given examples of the kinds of things they are and aren't allowed to say. Bing was soon fixed. One would think Microsoft would have preempted this problem before releasing such a high-profile product. Incredibly, the company's chief technology officer claimed that the problem would have been "impossible to discover in the lab."³

That would certainly not have been true with the right teams in place.

Generative AI has led to a bit of a gold rush, and technology companies have repeatedly cut ethical corners in their haste to release products. This was made easier by the fact that those companies had spent the previous two years eliminating or sidelining internal voices who could have told them to think twice and slow down. Notably, in 2021, Google fired two computer scientists on its AI ethics team, Timnit Gebru and Margaret Mitchell, after internal criticism of Google's approach to text-generation AI.⁴

Then there is the problem of misinformation. A New York lawyer used ChatGPT's help in preparing a legal brief, presumably oblivious to ChatGPT's disclaimers that it can generate inaccurate information. It made up a whole list of fake cases as precedents. The lawyer asked it if the cases were real, and the chatbot said yes—not having the ability to recognize that they were fake. It even made up entire judicial opinions. The lawyer submitted a brief based on its answers and, unsurprisingly, got into hot water with the judge.⁵ Attorneys facing penalties after submitting inaccurate briefs because of AI use has become a common occurrence.

Other examples of chatbot-related harms are more serious.⁶ Take companion chatbots, which are intended to provide emotional and mental health support. Amid an epidemic of loneliness, many people have come to rely on them. In fact, users of these bots, on average, reported that companion bots positively impacted the quality of their general social interactions, their relationships with family and friends, and their self-esteem.⁷ But the benefits are not uniform. A companion chatbot called Chai was implicated in the suicide of a Belgian man.⁸ Mentally unstable, he had become isolated from family and friends. He

used the app for six weeks and talked to it about his worries. After his death, chat records showed that the bot had encouraged him to kill himself.

Let's turn to the downsides of image generators. They have already had a major negative impact on many professions, notably stock photography. Why pay for a stock photograph of "Multiethnic Group of Friends Laughing Together" or "Healthy Organic Vegetables on a Wooden Table" when you can put those same prompts into an image generator and generate one—or ten—for free? Of course, image generators are trained on vast amounts of stock photography for which the photographers weren't compensated.

In general, AI reflects the biases and stereotypes captured in its training data. This is especially true of image generators. When MIT student Rona Wang wanted to create a LinkedIn profile photo, she asked an AI tool to make a casual photo of her look professional.⁹ It gave her lighter skin and blue eyes. Another example is Lensa, an app that creates stylized images from photos. When women uploaded their images, it generated sexualized images and nudes.¹⁰

Far worse than unintended sexualization is intentional sexualization. There is a community of programmers who use AI to generate pornographic images of various female celebrities, depicting all manner of sexual acts. The technology can just as well be used on unsuspecting everyday people.¹¹

What should we make of all this? Are these the growing pains of a nascent technology or is this inherent to the way it works? And as the technology advances, what new harms might arise?

Recall that we organized AI developers' claims about their products on a spectrum of truthfulness in figure 1.2. Some products do what it says on the tin. Others don't work at all. In

between those two extremes are products that are useful but oversold. Each of these can be harmful, in different ways. Generative AI is a mixed bag.

AI is good at certain things, which is also what can make it particularly harmful. For instance, AI is very good at generating stock photos, and that's exactly what makes it harmful to the stock photographers whose photos were used to train AI without compensation. But there are limitations to what generative AI can be realistically used for, and hype about chatbots and lack of awareness of these limits has led to serious problems. A lawyer can usefully use AI—in fact, even before generative AI, legal tech was a mature industry—but needs to use it discerningly. Finally, we also find some snake oil, such as a supposed “robot lawyer” product that can argue cases before the Supreme Court. AI can provide low-level assistance with filing legal briefs, but autonomously arguing a case effectively is far beyond its current reach.¹²

The varied landscape of generative AI applications resists a simple characterization of the limits of the technology. So, in order to evaluate claims about the technology's current or future capabilities—and to identify snake oil—it is necessary to understand how the tech itself works.

In this chapter, we will explore generative AI through the historical arc of the ideas that go into it. If your first exposure to generative AI was a chatbot like ChatGPT or an image generator like Midjourney, you might be surprised to know that this technology has an eighty-year history. It is in fact the result of a long series of gradual improvements.¹³

Retracing this path will help us build up our intuition about the utility and limitations of this technology one step at a time. It will also teach us something about the culture of the AI research community. And a big part of the payoff of this chapter

is psychological—once we understand the inner workings of generative AI and it no longer feels so mysterious, we will be better mentally equipped to resist the tendency to defer to claims made by those who built it.

To be clear, this chapter is far from a complete history of AI. There were many other forks in the road that could fill up entire books. We will limit ourselves to the advances that brought us to today's generative AI systems.

Let's dive in.

Generative AI Is Built on a Long Series of Innovations Dating Back Eighty Years

In 1943, neuroscientist Warren McCulloch and logician Walter Pitts published a mathematical model of how a neuron operates.¹⁴ The idea is simple: neurons are connected in the brain by synapses. Synapses are like wires. They carry an electrical signal in one direction from one neuron to another. A neuron fires—that is, generates an electrical signal—if there is enough signal coming into it through synapses.

Today we know that this model is vastly oversimplified. Still, the idea that the brain operates by calculating well-understood mathematical formulas—albeit on a scale of trillions of neurons—was inspiring to early AI researchers. They were eager to apply this idea to the pursuit of an intelligent machine, starting by building a mechanical version of a single neuron.

This was first achieved by Frank Rosenblatt, a psychologist, in the late 1950s. His team custom-built a computer, shown in figure 4.2, to implement what he called a perceptron.¹⁵ Designed to be the artificial equivalent of a single neuron, the perceptron could distinguish between two different shapes, say a triangle

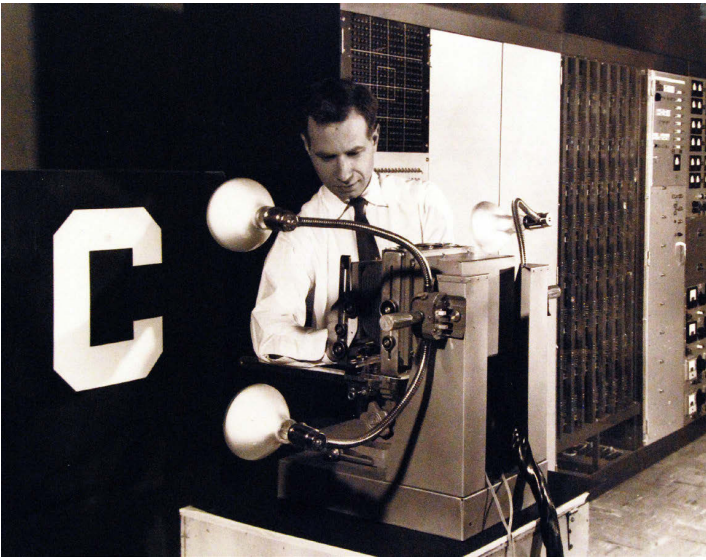


FIGURE 4.2. The camera system of the perceptron.

(Source: National Museum of the U.S. Navy—330-PSA-80-60 (USN 710739), Public Domain, <https://commons.wikimedia.org/w/index.php?curid=70710209>.)

and a square, or between two different letters. It had four hundred inputs, each input representing a pixel. Those four hundred pixels were arranged in a 20×20 image which was generated by a primitive digital camera.

The reason the perceptron was exciting was that it could learn to classify images without having the shapes of letters manually programmed into it. It classified images based on the strengths of the connections between each pixel and the output unit (these strengths are called “weights” in AI). If you trained the model by feeding it a bunch of pictures of “A” and a bunch of pictures of “E,” say, and told it which pictures corresponded to which letter, it would automatically adjust the strengths of its connections accordingly, so that it could then correctly classify

any A or E image it was presented with. The perceptron was one of the first machine learning systems ever built (a checkers-playing program that learned from its mistakes had been built a few years prior¹⁶).

Note that the machine could do only binary classification, that is, distinguish between two different shapes. If you wanted to use it to classify an image as one of the twenty-six letters of the English alphabet, you'd have had to make it substantially more complicated.

A Rosenblatt perceptron can be thought of as a sequence of four hundred numbers representing the weights or connection strengths. Those numbers are generated as part of the learning or training process, and together they completely determine its functionality. If we wrote them out on a piece of paper, anyone could use that information to build a machine that would work exactly the same way.

Today's neural networks can also be described as a long sequence of numbers. But for the largest models, that sequence would be over a billion times longer than the perceptron's. If you printed it all out, it would make a stack of paper hundreds of miles high. AI has come a long way. Let's take a look at how that happened.

Failure and Revival

Just as a single neuron in the brain can't do much in isolation, a single perceptron isn't particularly useful. So researchers started looking into building bigger neural networks, with neurons arranged in "layers," one layer feeding into the next.

These multilayer networks were in a sense before their time, and they exposed the limitations of computing power in the mid-twentieth century. For example, fully connecting each

layer to the next layer was computationally expensive. In this design, if there are one thousand neurons per layer, every neuron in one layer would need to be connected to every neuron in the next layer, resulting in a million connections per layer. The hardware at the time was not capable of running these networks, so researchers instead focused on networks with more limited connections.¹⁷ But these networks were nowhere near as capable as fully connected networks would have been.

As a result, interest in neural networks faded in the 1970s, and the AI research community's attention turned to symbolic systems. Symbolic systems are a very different way to build AI, and they have a parallel history to neural networks, which we won't dive into. But it is worth understanding their basics. While neural networks operate on numbers—such as by multiplying and adding them together—symbolic systems operate on discrete symbols or categories. While neural networks learn from data, symbolic systems have their rules programmed into them. While neural networks see statistical patterns, symbolic systems do calculations or use logical reasoning. One example of a symbolic system is a chess-playing computer that examines billions of possibilities of the form “If I play X, my opponent might play Y, and then I could play Z, and I would be winning.”

The change in the relative interest in neural networks and symbolic systems was the first in a repeated pattern. When the AI community becomes excited about a particular approach, a feedback loop is created in which researchers and funders influence each other to propel work in that area forward. Peer reviewers, who have a big hand in deciding what research is published, are often skeptical of research on approaches that have fallen out of fashion. As a result, the field often moves in sync toward the approach of the moment, almost entirely abandon-

ing earlier research programs. This is what happened to neural networks research in the 1970s (and in fact, symbolic systems are in such a position today).

Since the 1970s, neural networks have ridden several waves of popularity. The first revival came in the 1980s, when researchers established two important and related ideas that reinvigorated neural network research. First, it helps to make your network deep in the sense of having many layers, as shown in figure 4.3, if computational capacity allows. That's because the stacking of layers allows the network to learn increasingly complex concepts. (We'll see why in a moment.) Second, a clever algorithm called backpropagation can train deep neural networks. Though these ideas have a long history, they came together in a paper in the journal *Nature* in 1986.¹⁸ One of the authors was Geoffrey Hinton, later recognized with a Turing Award (often referred to as the Nobel Prize for computer science) for his contributions to AI.

But in the 1990s, neural networks faced another period of declining interest. They were supplanted by a different technique called SVM, which stands for Support Vector Machine. Despite the name, SVM refers to an algorithm, not a physical machine.

SVMs became popular because they were more computationally efficient than neural networks and therefore could be run on cheaper hardware. For example, since the 1980s, the United States Postal Service had been using digit recognition machines to automatically sort mail by zip code.^{19,20} A few years later, SVMs were shown to be as or more accurate than neural networks at digit recognition, while also being more efficient.²¹ This demonstration had a big impact on machine learning researchers.

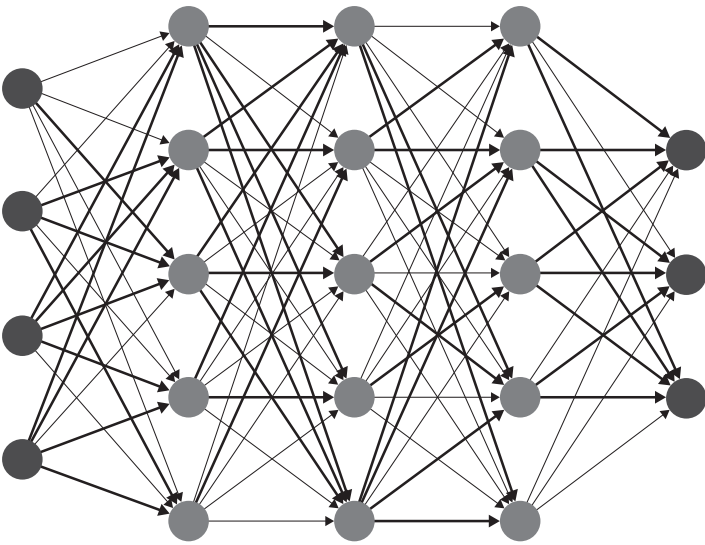


FIGURE 4.3. A five-layer neural network with random weights (represented by the thickness of the connections).

Today, the computational demands of handwritten digit recognition are trivial, so SVMs have no real advantage. And on the more complex image processing tasks that we'll discuss in the next section, SVMs don't work well at all. But the kinds of tasks that would show the superiority of neural networks were simply not on the radar back then. Besides, SVMs came with neat mathematical theory that explained why they worked, which appealed to researchers. Neural networks, in contrast, were a bit of a black box. The question of how important it is to understand why learning algorithms work was—and still is—a debated topic in AI, as is the related question, Is it more important to have theoretical understanding (based on mathematical proofs) or empirical understanding (based on experimental evidence)?

Training Machines to “See”

In 2007, the field of computer vision was somewhat stagnant. Fei-Fei Li, a new professor at Princeton, believed that the bottleneck wasn’t due to a lack of clever machine learning models, but rather to a lack of available data to train those models. She reasoned that more data would settle long-standing debates about what techniques were best for computer vision: neural networks, SVMs, or other techniques. She assembled a team to collect a big dataset of images from the web, which they called ImageNet.²² Initially, her ideas were scoffed at and the project received little funding, yet the team pressed on.

Li needed to hire people to manually label lots of images—whether objects like “bike” and “cat” or concepts like “anger”—so that AI could learn what those objects or concepts looked like visually. This task was going to be hard on a shoestring budget. By chance, she learned of Amazon Mechanical Turk, a website that enables people from all around the world to complete small online tasks for pennies.²³ This cheaper approach to data annotation made the project possible.

In 2009, Li and her graduate students, by then having moved to Stanford, launched ImageNet publicly. At the time, it had about three million images arranged into over five thousand categories, and it would eventually grow to over ten million images.

ImageNet didn’t make much of a splash. But the following year, led by graduate students Olga Russakovsky and Jia Deng, the team launched a competition to see which AI model could most accurately classify images.²⁴ They selected a subset of ImageNet consisting of about a million images and designated it as the training dataset. Any researcher could participate by training a model on those images, and the models would be

ranked by how accurately they classified images in another designated set of one hundred thousand images.

Running competitions like this is a time-tested tradition in AI development. It is extremely easy for AI researchers to fool themselves into thinking that a model is more capable than it actually is. For instance, they might evaluate it using data that is particularly easy for the model to classify. Or they might evaluate how well the model works using the data that was used to train it—like teaching to the test. Through these competitions, researchers can ensure that everyone uses the same training data and that the test data is kept secret. The result is a leaderboard that represents a fair test of models' capabilities.

It is this practice of benchmarking that has historically allowed rapid AI development. To evaluate a new proposed idea for improving AI, researchers don't need to carefully analyze it to form an opinion, much less wait for the cumbersome peer review process to weigh in. They can simply look at whether the idea results in an improvement in the state-of-the-art accuracy when used on one or more benchmark datasets. Further, the tweaks to models that different researchers propose can often be combined without much effort.

In the first two years of the ImageNet competition, the winning models—which were based on SVMs—had high error rates, misclassifying over a quarter of the images. Having such high error rates meant that these models could not be used in practice, for instance, to automatically label photos to enable later searching in a camera app.

In 2011, Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton decided to take a crack at the ImageNet competition using neural networks, which by then had been branded “deep learning” because of the key insight that having more layers (depth) im-

proves accuracy.²⁵ Krizhevsky was a grad student who would later work at Google. Sutskever would go on to cofound OpenAI. Hinton, the aforementioned Turing Award winner, had published the *Nature* paper on backpropagation twenty-five years previously.

It took them a year to develop their neural network, which they named AlexNet.²⁶ They managed to push the number of layers to eight, almost unprecedented at the time, using a new technique that made it possible to train deeper networks. All of this required massive computational power, but fortunately for them, such power had just become available. A few years prior, hardware companies had figured out how to repurpose the extremely powerful but highly specialized graphics processors (GPUs) used in video games. In 2011, the year work on AlexNet started, researchers had for the first time published details on how to use GPUs for fast AI training.²⁷ The AlexNet team was able to build on this approach.

They entered AlexNet into the 2012 contest. It won by a massive margin: 85 percent accurate compared to 74 percent, a difference unheard of in a game of inches. It immediately became obvious to researchers that deep learning would be the way to go in computer vision, overcoming the prevailing skepticism of the approach. In that moment, the field of AI changed permanently. A massive influx of researchers raced to tweak these new deep-learning-based models, and three years later the state-of-the-art accuracy surpassed 96 percent, enabling a slew of never-before-possible practical applications.

Soon the algorithms got good enough that the depth—the number of layers—was no longer constrained by computing power and could be arbitrarily high. By 2015, some models had well over a hundred layers.

The Technical and Cultural Significance of ImageNet

ImageNet had a huge impact on AI research, even beyond computer vision. First, it showed that no other known technique can compete with deep neural networks for perception tasks. Around the same time, deep learning outperformed traditional methods on speech recognition tasks; its success in computer vision showed that that was no coincidence.²⁸ The use of other machine learning methods like SVMs in computer vision quickly disappeared. Within a few years, deep learning came to be seen as the approach of choice for an expanding set of machine learning applications, especially those where large datasets were readily available. For example, many of the algorithms behind social media news feeds use deep learning. They are trained on our collective behavior, such as what we click on or comment on.

In cases where datasets were not readily available, ImageNet showed the value of assembling such datasets. Since the ImageNet era, dataset sizes have only increased. In 2017, Google revealed that it internally used a dataset of three hundred million images for training models; that's three hundred times larger than the training data for the ImageNet contest.²⁹ Many chatbots are trained on a dataset called the Common Crawl, which is a collection of over three billion web pages totaling trillions of words.³⁰

ImageNet also made clear that graphics processors are essential for training deep neural networks. Companies such as NVIDIA, a manufacturer of graphics processors, benefited immensely from this boom. In 2023, it became a trillion-dollar company. These days, the vast majority of AI computations, both in data centers and on consumer devices, take place on dedicated chips that are very similar to graphics processors but

are more optimized for neural networks. For example, the chip used in the iPhone 13 Pro, released in 2021, can perform about sixteen trillion—that's sixteen million million—arithmetic calculations per second.³¹ The racks of computers used in data centers are, of course, thousands of times more powerful.

At least as important as the technical significance of ImageNet is the way it shaped the culture of AI research and development. It either gave rise to or contributed to many modern AI practices.

Web scraping has become the standard way to collect text or images from the internet for AI training. Scraping is done by bots, or automated web browsers, that record the contents of billions of web pages, and in some cases (like the ImageNet competition), this data is then labeled by humans.

The data annotators on Amazon Mechanical Turk were typically paid low wages, about two dollars per hour.³² Further, ImageNet developers did not compensate the photographers who created the images, most of whom did not know that their work would end up as training data. ImageNet operated on a small budget and might not have existed if not for this way of gathering and annotating data. But even today, commercial projects by trillion-dollar companies operate the same way.

One downside of scraped datasets is that it is hard to manually examine them for problematic content. ImageNet contained numerous slurs and offensive labels, as well as pornographic images of people who did not consent to their inclusion in the dataset.³³ The ImageNet team released filtered versions much later, in 2020 and 2021.^{34,35}

These lax practices were taken up by companies, and as a result, their products sometimes behave in unintended ways. In 2015, a user of Google Photos discovered that the app tagged a

photo of him and a friend, who are both Black, as “Gorillas.” In response, Google and Apple simply prevented their photos apps from ever producing that label, even for pictures of actual gorillas.³⁶ Presumably, fixing the training datasets was not considered an option. Eight years later, you still can’t search for gorillas—or most other primates—on these apps.

Even before ImageNet, nearly the only way to get an AI innovation taken seriously by the research community was to achieve state-of-the-art results on a benchmark dataset. ImageNet further entrenched this norm. As we mentioned earlier, benchmarking does allow rapid progress—but it is a one-dimensional kind of progress that may not represent what we want out of AI in the real world. For example, most benchmarks don’t measure the extent to which models reflect cultural biases and stereotypes. AI engineers, meanwhile, have long complained that benchmark-beating models are too complex, and hence too slow and brittle to use in real apps.

ImageNet provided further evidence of the long-standing observation in AI that methods that relied on encoding expert knowledge were eventually outperformed by methods that relied on the machine discovering that knowledge from data. As early as 1985, renowned natural language processing researcher Frederick Jelinek said, “Every time I fire a linguist the performance of the speech recognizer goes up,” the idea being that the presence of experts hindered rather than helped the effort to develop an accurate model.³⁷ Prior to the use of deep learning, computer vision researchers hand-coded algorithms for converting pixels into conceptually meaningful elements of images. All of that proved unnecessary and even counterproductive. But, yet again, the absence of experts also makes it harder to assess the behavior of AI systems on dimensions not captured by the benchmarks.

Something else is lost in this quest for general-purpose methods that don't require human expertise: when the model is not tailored to the task at hand, it might require many more training examples to reach a given level of accuracy. But having established a culture where existing data can be appropriated for machine learning, this did not seem like much of a barrier in the AI community.

In deep learning, researchers use a single algorithm across the board, with minor variations. It's called gradient descent, and the idea is basically to tweak the weights (connections between neurons) by a tiny amount every time the model makes a mistake. To be sure, small improvements to this algorithm are frequently discovered. But the effect of those improvements is mainly to make training faster, not to enable new applications. It's incredibly empowering to AI developers that they don't have to design a new algorithm whenever they want to tackle a new task. It may be hard to believe, but the same learning algorithm is behind chatbots, text-to-image generators, and thousands of other AI apps that perform varied functions. What differs is the training data and the architecture (the broad pattern of connections between neurons).

Finally, ImageNet continued the culture of open knowledge sharing in AI research. It was a great demonstration of how quickly progress could happen if researchers built on each other's work. Even though many of the contestants worked for companies who were in competition with each other, the norm of openness held sway. If one company decided not to publish its methods, AI researchers would find it less appealing to work there, because they wanted their discoveries to contribute to human knowledge and not just to a company's bottom line. This would put such a company at a competitive disadvantage. Today, this culture has changed to some extent as companies

prioritize profits. The question of whether AI knowledge should be shared or hidden has become a major flash point in the community.

Classifying and Generating Images

Let's take a minute to understand why deep neural networks work so well for image classification and generation. Consider the photograph of a dog playing in a park, shown in figure 4.4.

If you zoom in far enough on this photo on a computer, you'll see that it consists of pixels—colored dots. But on zooming out a bit, these dots turn into patterns, shapes, and objects. You might notice the color contrast between the dog and the background, the texture of the fur, or the shape of the ball in its mouth.

A deep neural network is trained to discover these types of visual concepts based on how the pixels are arranged. Crucially, the layers of the network encode successively more complex concepts. The concepts in each layer build on those in the layer that precedes it.

We can see this concept in the images in figure 4.5, made available by Google researchers.³⁸ It is a deep neural network trained on ImageNet. Each image shows the input pattern that the corresponding neuron learns to detect. Earlier layers focus on simple concepts such as edges, textures, and patterns, whereas later layers are able to detect objects' parts and, finally, specific objects, like the dog in the photo in figure 4.4.

This understanding helps explain a shocking discovery that came out of the ImageNet contest: once a model is trained to classify images, it can be adapted to a variety of visual tasks with relatively little effort, through a process called fine-tuning.

As just one example, consider reverse image search—that is, using one image to find many other similar images on the internet. This is amazingly useful. When you're out on a walk, you can

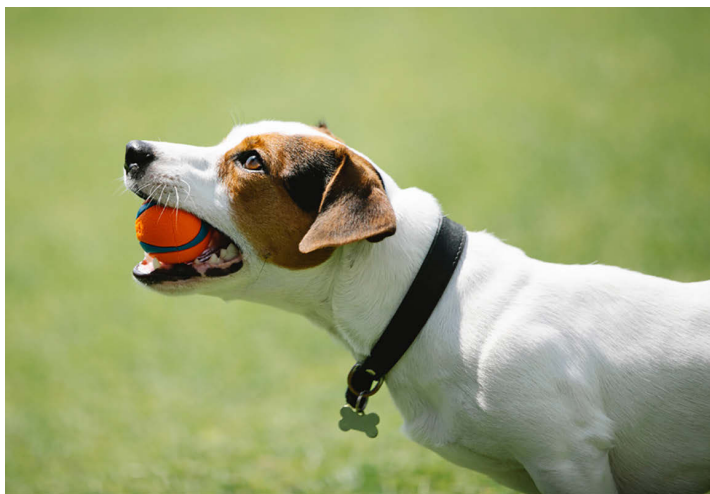


FIGURE 4.4. How do neural networks classify images of dogs?
(Photo by Blue Bird: <https://www.pexels.com/photo/dog-in-black-collar-with-ball-in-teeth-7210262/>.)

snap a picture of a tree with your phone and instantly learn which species it is.

It turns out that the hard part of image classification is getting a model to learn the visual structure of the world—its patterns, objects, and so on. This is what a deep learning model uses most of its layers for—all but the final layer. The final layer is a simple process of converting those concepts to labels—the words or categories we use to describe images.

At the penultimate layer, the model outputs a sequence of numbers, called a vector, that corresponds to a high-level description of the image. As shown in figure 4.6, two images of dogs will result in sequences of numbers that are close to each other. Two images of Pomeranians will be even closer, and images of the same Pomeranian closer still. So, if you start with an image of a Pomeranian, convert it into a sequence of numbers, and

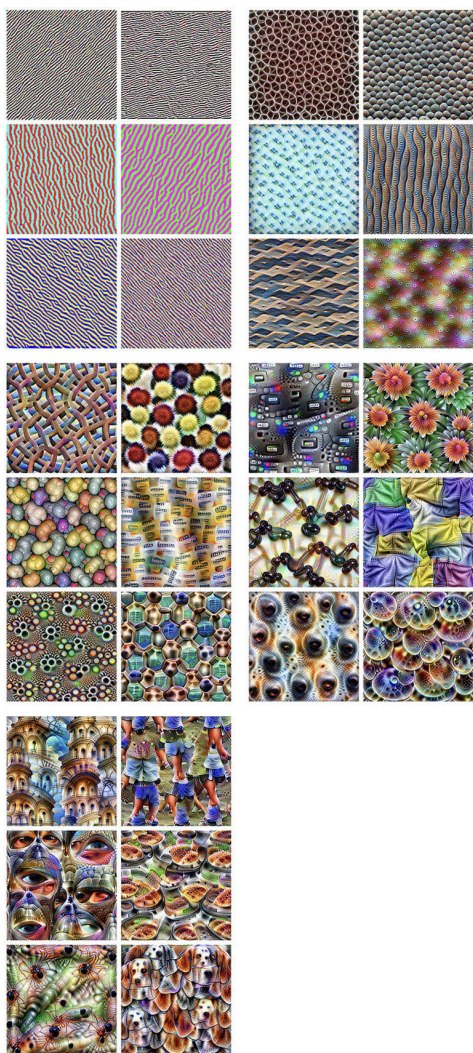


FIGURE 4.5. Visualization of some of the concepts learned by a deep neural network trained to perform image classification: edges (left), textures, patterns, parts, objects (right).

(Photo by Olah et al., “Feature Visualization.” Distill, 2017.

[https://distill.pub/2017/feature-visualization/.](https://distill.pub/2017/feature-visualization/))



FIGURE 4.6. Illustration of vector similarity, with four sequences of sixteen numbers visualized through color coding. A is 98 percent similar to B, 90 percent similar to C, and unrelated to D. Hypothetically, A and B might result from images of the same dog, C from another dog, and D from, say, a living room.

look for similar sequences in a database, you will find other Pomeranians. It's a stunningly simple approach.

So far, we have talked about how images can be classified. But tools like Dall-E and Stable Diffusion allow us to generate images, not just classify them. How do they work?

The dominant type of text-to-image technology in use today is the diffusion model. These models learn how to gradually transform an image consisting entirely of random pixels, called noise, into a structured image. The key insight is that if you keep adding noise to an image, it will eventually become unrecognizable—like static on a TV screen. A model can then learn the *reverse* process of converting noise back into recognizable images, given a caption for the image, as shown in figure 4.7.

This transformation is guided by a trained neural network. The model has learned, typically using a large dataset of images and their captions, how to progressively shape this noise into recognizable forms. At each step, the model makes predictions about what the final image should look like, based on its training and the caption of the image. It then adjusts the current noisy image to move closer to that prediction. This is done iteratively, with each step refining the image further.

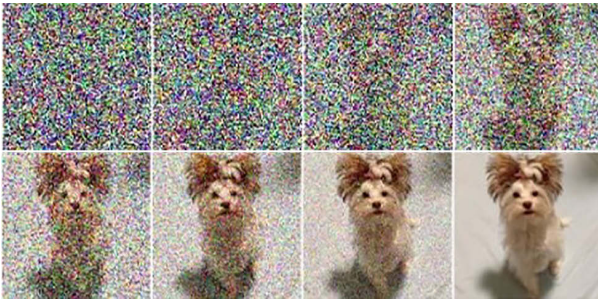


FIGURE 4.7. Generating an image of a dog using a diffusion model. (Photo from Song Y, “Generative Modeling by Estimating Gradients of the Data Distribution” (blog). May 5, 2021. <https://yang-song.net/blog/2021/score/>.)

In order to be useful, text-to-image models need a large amount of data. As we have seen, one of the key insights of the ImageNet challenge was that using a dataset with over a million images could qualitatively improve the resulting models. This is doubly true for generative models. The dataset used for training Stable Diffusion contains billions of images, a thousandfold increase over the ImageNet dataset.

With this understanding of image classification and generation, we are in a better position to understand the harms that arise from the ways in which these technologies are built and sometimes (ab)used. In the next two sections, we will make a brief detour from our technical exposition to discuss these aspects.

Generative AI Appropriates Creative Labor

The success of all generative AI depends on the availability of a large amount of data. Stability AI used a dataset of over five billion annotated images scraped from the internet to build the text-to-image tool Stable Diffusion. But the artists who created this

training data were never asked for consent.³⁹ Companies claim that their use of images as training data falls under the fair use provision of U.S. copyright law, which allows creators to use copyrighted materials without permission in some circumstances.⁴⁰ But these laws were last revised in 1978. At the time, an application that could automatically generate passable images and text could not be imagined, let alone prohibited by law.

The legal and ethical questions are even more acute when you consider the fact that image generators allow you to generate an image in the style of a particular artist.

Generative AI models can also “memorize” their training data. That is, they can output near-exact copies of images and text in their training data. Images generated using Dall-E and Stable Diffusion occasionally include watermarks from stock image websites such as Shutterstock and Getty Images, showing how prevalent watermarked images are in the data and how easily the model can replicate parts of its training data.

You can test memorization by prompting an image generator to produce any famous painting. Figure 4.8 shows a striking example. As image generation technology advances, its ability to reproduce images of paintings has been getting more accurate. In some cases, though not always, even relatively obscure images in the training data have been found to be memorized.

Technical measures are being developed to make it less likely for generative AI tools to output copies of items in their training data. But some models like Stable Diffusion have been openly released and have already been downloaded hundreds of thousands of times. Once a model is made available for downloading on the internet, it is nearly impossible to restrict how it is used.

The ramifications are serious. Artists and photographers have had their work and style copied without compensation.^{41,42}

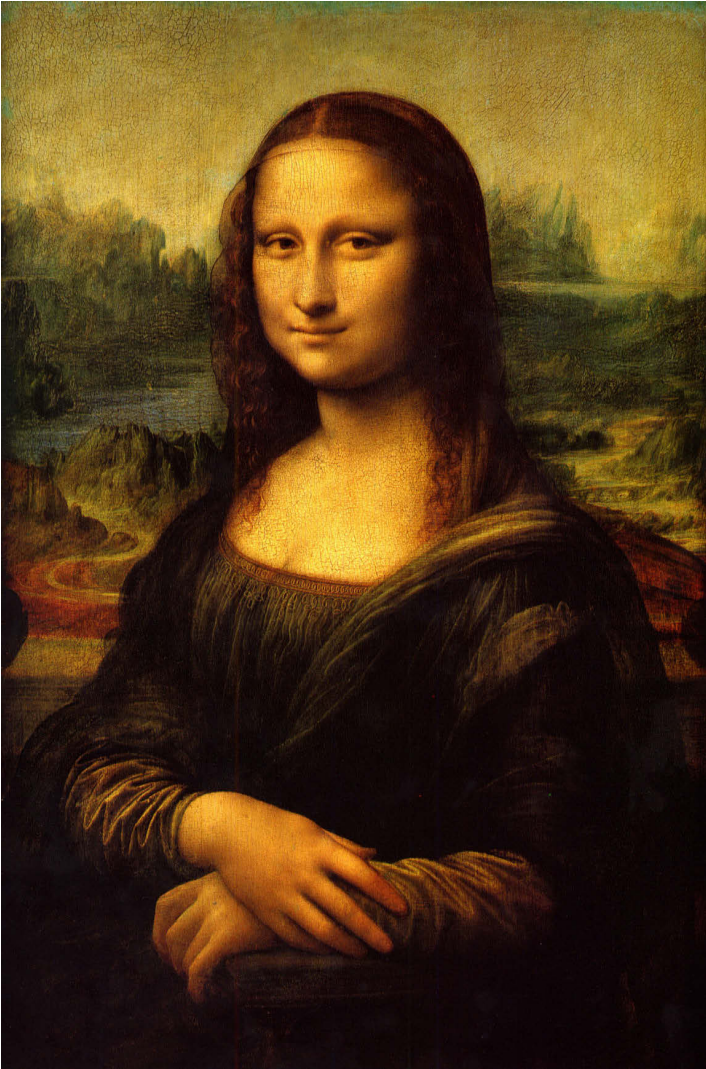


FIGURE 4.8. Text-to-image tool Midjourney replicates the Mona Lisa.
(Sources: *Left:* Original, by Leonardo da Vinci—Musée du Louvre, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=51033>. *Right:* Midjourney.)

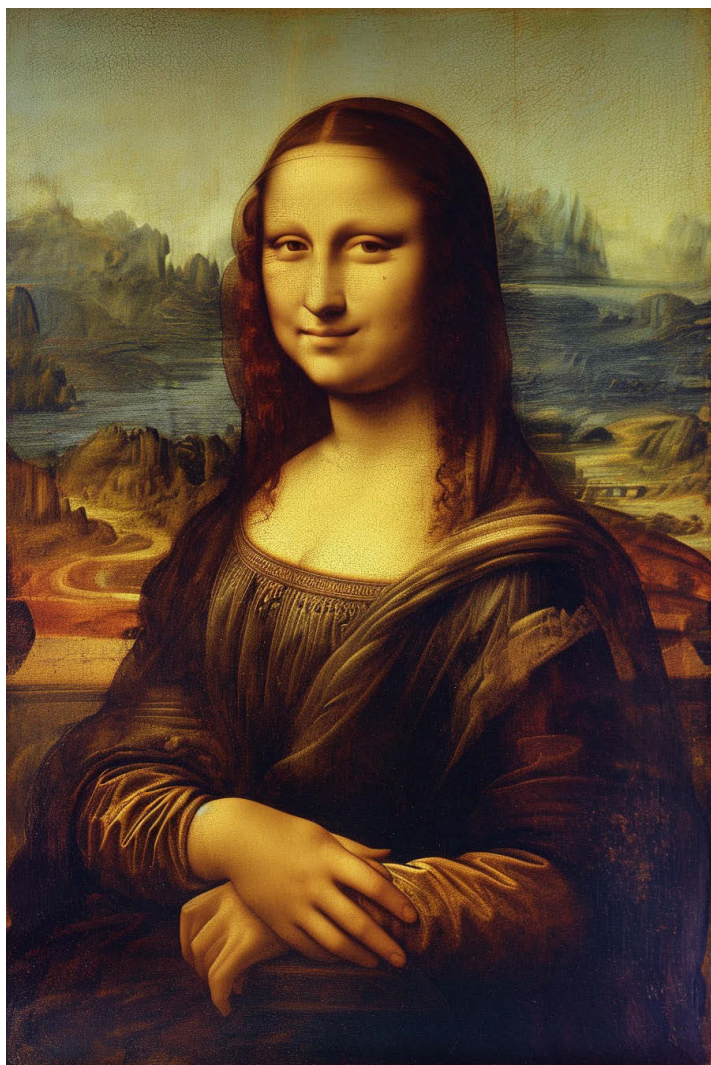


FIGURE 4.8. (*continued*)

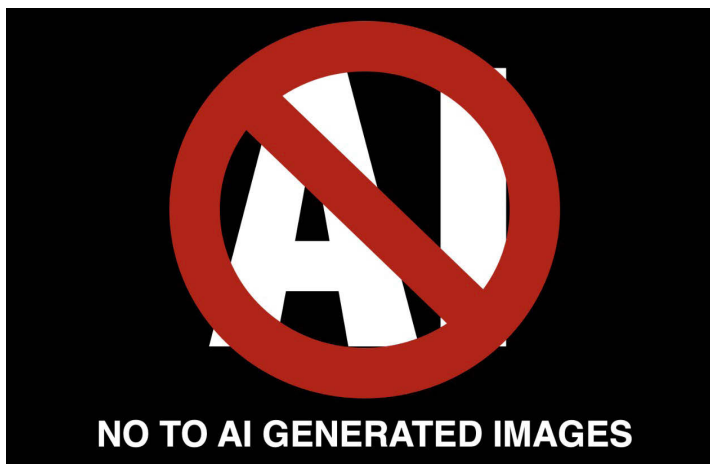


FIGURE 4.9. ArtStation was flooded by similar images uploaded by artists protesting generative AI.

Sometimes the generated images even include remnants of artist signatures.⁴³ Creators see none of the profits. The fear is that these tools will be used to replace rather than augment human artists,⁴⁴ especially for routine commissions like cover images and company logos. And if we overwhelmingly replace artists with generative AI, whose data will train the next generation of AI models?

Creators are fighting back. In December 2022, ArtStation, an online artist community, saw a widespread community protest. Artists stopped uploading original images and spammed the website with the slogan shown in figure 4.9: “No to AI Images.”⁴⁵ Many artists want developers of generative AI to find a way to work with them—rather than simply using their work without compensation, consent, and credit—while others see the technology as irredeemably harmful to artists, art, and culture.

AI for Image Classification Can Quickly Become AI for Surveillance

Unlike predictive AI, which is dangerous because it doesn't work, AI for image classification is dangerous precisely because it works so well. The same technology that can be used for image classification and reverse image search also allows for mass surveillance. Image classification and face recognition are technically very similar.

Problematic uses of government uses of facial recognition abound. The South Korean government handed over 120 million photos of visitors to the country to a private company to develop a face recognition system.^{46,47} In 2021, the state of Telangana in India faced a lawsuit because police photographed people without consent or explanation during routine traffic stops.⁴⁸ Telangana has the highest number of face recognition tools being built of any state in India, and people suspect their photos will be used as training data for these systems. Several Chinese startups sell face recognition systems to the government. They proclaim the effectiveness of identifying the minority Uyghur population based on images.⁴⁹

Clearview AI is a company that sells face recognition technology and has faced criticism for the way it built its technology and the way it has been used.^{50,51} Clearview collected twenty billion images of people from social media to create a database.⁵² Many misuses of Clearview's product have been reported. The company's own employees used the face recognition app on unsuspecting people to track where they had been and what they did for a living. Unbeknownst to the public, police officers in municipalities across the United States used Clearview's app. Officers used the tool even when their department had not

approved it, and they used it for reasons unknown to the public or their supervisors, showing the lax enforcement of rules on face recognition.⁵³

Facial analysis technology is also being used on physical billboards. Instead of a billboard displaying a constant or a rotating set of advertisements, companies are using AI to recognize the characteristics of people looking at a billboard, such as their age and gender, and changing the advertised product to tailor it to their demographic.^{54,55} It is not hard to imagine this going a step further; companies could use face recognition to identify each passerby and tailor advertisements based on their personal information and interests. Each of us might see a different advertisement when we look at a billboard—like the online advertisements we get from Facebook and Google. If this sounds right out of a dystopia, that’s because it is—personalized physical billboards appear prominently in the 2002 film *Minority Report*.

We expect companies to continue developing surveillance tools. After all, it’s good for business—Clearview AI is valued at over USD 100 million. But public pressure, advocacy, and regulation can curb how AI is used for surveillance.

When Clearview’s abuse of publicly available data came to light, privacy regulators in several countries started investigating the company. They asked Clearview to delete citizens’ face recognition data in Italy, France, Australia, and the UK. France and Italy each imposed a fine of EUR 20 million, and the UK imposed a fine of GBP 7.5 million. Canada’s privacy regulator opened an investigation in 2020, as a result of which the company stopped selling its face recognition tools in Canada. In 2022, the country’s privacy commissioner recommended a moratorium on using face recognition by police and private

industries. Stronger regulations and more public pressure will go a long way in curbing AI for surveillance.

Challenges from community organizers, activists, and civil rights advocates have proven successful in getting the attention of policymakers. Over a dozen U.S. municipalities have banned face recognition after widespread public demands by advocacy groups and community organizations. We'll come back to the potential of regulation to curb AI harms in the last chapter.

From Images to Text

Let's now return to our technical discussion of generative AI, turning to text generation.

To recap, deep learning makes it possible to apply the same learning algorithm to many different tasks. If we look at the code that is used to build an image classifier using ImageNet, relatively little of it is specific to the fact that what's being classified is images. The fact that the weights of the trained model correspond to visual concepts is a consequence of the training data, not the algorithm.

So what would happen if we trained the model by feeding it text instead of images? We could give it a bunch of news articles, each labeled by topic. Once trained, would it then be able to classify new articles by topic? Similarly, could we train other useful tools, say one that determines whether a social media post about a movie expresses positive or negative feelings? That would be an interesting way to build a barometer of public opinion about a movie on its opening weekend. Or perhaps we could train a classifier to determine whether a piece of text is intended to be funny, which has traditionally been a notoriously hard problem in AI?

One obvious difference is that images are made up of pixels while text is made up of characters. But that's actually a relatively superficial difference, and we could tweak the code to account for it. A much bigger problem is that the structure of images is different from that of text. Images have a strong spatial structure; each pixel is strongly correlated with those near it. Text is different. Words are related to adjacent words, but there are also so-called long-range dependencies. To understand what that means, consider this classic joke:

According to unofficial sources, a new simplified income tax form contains only four lines:

1. What was your income for the year?
2. What were your expenses?
3. How much have you left?
4. Send it in.

For the punchline at the end to make sense, you have to remember that the joke is about income tax, which was mentioned near the beginning. The correct interpretation of a sentence might hinge on one or two words that were mentioned many paragraphs earlier.

Images are less likely to have such long-range dependencies. A pixel in one corner of the image is unlikely to change how we make sense of another corner of it.

Researchers have tried many ways to capture this long-range structure in text. But until the late 2010s, it remained a big challenge. Early automated language translation apps worked well for short sentences but ran into trouble with longer ones. For an app such as a chatbot that must remember the

context of the conversation across many, many sentences, this was a big problem.

A breakthrough came from Google in 2017. It was a brute-force solution involving computing a big matrix—a grid of numbers—when processing a piece of text. The matrix quantified the degree to which each word in the text was related to every other word. Processing big matrices was just the kind of task at which GPUs excelled, because it involved performing a large number of calculations in parallel. By 2017, GPUs had become ubiquitous in AI.

Using matrices to capture aspects of structure in language, neural networks are able to build up successively more complex concepts as information flows through their layers, just as they do with images. Suppose the input being processed is a story. The lower layers of the network might track simple relationships such as the connection between a noun and its corresponding verb. As the network “reads through” the story, higher layers may keep track of information about characters, such as relationships, locations, personalities, and goals.

Going back to the income tax joke, lower layers won’t make the connection between the subject of the joke and its punchline, but higher layers will. Today, ChatGPT has no problem explaining this joke and others like it. In fact, this matrix was the main innovation that enabled an architecture called the Transformer, which is the *T* in ChatGPT.

But wait a minute. So far, we’ve talked about how you can build a neural network to *classify* a piece of text. How does ChatGPT actually *generate* text?

It turns out that going from text classification to text generation relies on another mind-bendingly brute-force method. Try to think of the simplest possible text classification task. It might

be this: given a sequence of words from a text, guess the next word in that sequence. In other words, autocomplete.

Next-word prediction is a bit different from the tasks we've looked at so far because there is no single correct answer. The same sequence of words, say "Once upon a time," may have different continuations in different texts. But it turns out that this doesn't matter much.

What does matter, and what makes the next-word-prediction task appealing to AI researchers, is that there is an astronomical amount of training data online. Unlike the tasks we've discussed so far that require specific training data, such as training a model to classify a news article by topic, literally any text can be used for training a neural network to predict words in a sequence. That includes the words you're reading now.

Even better, the training data doesn't need to be labeled—such as object labels in the case of ImageNet or positive/negative/neutral labels in the case of movie reviews. The sequence of words itself forms the labels. It avoids the need for what would otherwise be an unimaginably laborious manual annotation process.

So here's the big reveal—all modern chatbots are actually trained simply to predict the next word in a sequence of words. They generate text by repeatedly producing one word at a time. For technical reasons, they generate a "token" at a time, tokens being chunks of words that are shorter than words but longer than individual letters. They string these tokens together to generate text.

When a chatbot begins to respond to you, it has no coherent picture of the overall response it's about to produce. It instead performs an absurdly large number of calculations to determine what the first word in the response should be. After it has

output—say, a hundred words—it decides what word would make the most sense given your prompt together with the first hundred words that it has generated so far.

This is, of course, a way of producing text that’s utterly unlike human speech. Even when we understand perfectly well how and why a chatbot works, it can remain mind-boggling that it works at all.

Again, we cannot stress enough how computationally expensive all this is. To generate a single token—part of a word—ChatGPT has to perform roughly a *trillion* arithmetic operations. If you asked it to generate a poem that ended up having about a thousand tokens (i.e., a few hundred words), it would have required about a *quadrillion* calculations—a million billion. To appreciate the magnitude of that number, if every single person in the world together performed arithmetic at the rate of one calculation per minute, eight hours a day, a quadrillion calculations would take about a year. All that to generate one single response.

It is this sledgehammer of an algorithm that’s the core of text generation. It is the *G* in ChatGPT, which stands for “generative.”

From Models to Chatbots

The techniques we’ve described so far constitute what’s called a base model, which was state of the art in the late 2010s. In 2019, models like T5 from Google and GPT-2 from OpenAI were released. The capabilities of these models were clear to researchers, so they caused huge excitement in the AI community. But they weren’t useful as consumer products yet, so the public heard little to nothing of them. They were very good

models of human language but functioned more as a fancy autocomplete than a chatbot. If you tried to use them as a chatbot, you would be underwhelmed. Suppose you said:

What is the world's tallest mountain

It probably won't give you the answer. Instead, it might respond with

?," asked the teacher.

That's a perfectly good way to autocomplete that phrase! (The sentence that the model is trying to autocomplete being "‘What is the world's tallest mountain?’ asked the teacher.")

To get such models to do useful things, people used creative prompts like:

Q: *What is the world's tallest mountain?*

A:

Here the model thinks it is autocompleting a question-answer pair. Sometimes this worked, sometimes it didn't.

In the late 2010s, if researchers wanted to use a language model for a task such as translation, they attempted to fine-tune an existing model for that task, just as an ImageNet classifier can be fine-tuned for alternative tasks such as image search.^{56,57} This worked pretty well. Researchers would start with the base model and give it a few thousand pairs of sentences in, say, English and French. This would result in a translation tool.

This way of building a translation tool is far more effective than training one from scratch, because it requires orders of magnitude less data in the form of English/French pairs. Similarly, one could build a movie review sentiment classifier or a joke detector or a poem generator or a puzzle solver with relatively little annotated or labeled data.

This is a big win, because creating high-quality labeled data is expensive, whereas the unlabeled data used for pretraining the base model is essentially available for free online (but again, only because AI companies appropriate the labor of those who generated the text, without credit or compensation).

The reason fine-tuning is so successful is that even though the base model just does autocomplete, it already has language translation and all those other capabilities embedded within it. Why is this? Well, out there in the morass of the internet are millions of instances of sentences translated between languages. The model—primarily by virtue of being incomprehensibly large in terms of the number of neurons and connections between them—has learned the patterns that enable language translation simply through the process of getting better at guessing the next word in a sequence.

Fine-tuning merely changes the model's behavior; it “unlocks” specific capabilities. In other words, fine-tuning is an elaborate way of telling the model what the user wants it to do. But pretraining, rather than fine-tuning, is what gives it the capability to function in that way. This explains the *P* in ChatGPT, which stands for “pretrained.”

But we've still not built a chatbot. Using fine-tuning for language translation or other tasks requires programming, which is a whole different ball game from simply asking the bot to do the task. Knowing this, researchers tried to fine-tune the model to instead follow instructions. In other words, instead of fine-tuning it for translation or puzzle solving or any other specific task, they fine-tuned it for a meta task: “Follow the user's instructions.”

This was first convincingly demonstrated by Google researchers in late 2021.⁵⁸ Meanwhile, OpenAI researchers tried the same thing but used a slightly different type of fine-tuning,

which they described publicly in early 2022.⁵⁹ There are many steps involved, but the basic idea is to give it a bunch of labeled examples of commands and desired outputs (if the user says this, then you should say that).

That was the final innovation that led to ChatGPT. The shocking thing about this, even to machine learning researchers, was how well the bot was able to follow instructions. It was able to carry out tasks even if they weren't specifically present in the fine-tuning corpus, as long as the task was within the general set of capabilities the model had acquired through pretraining.

One limitation of current chatbots is that even though they are quite general, their abilities are limited to tasks that are somewhat similar to tasks represented in the training data. For example, GPT-4 plays passable chess, a remarkable fact considering that its ability almost certainly comes not from explicit instruction but simply from learning the statistical patterns in the hundreds of millions of games available online.⁶⁰ But the bot wouldn't be any good at a similar board game whose game transcripts haven't been posted online.

Nor can chatbots immediately learn a new task by interacting with you in the way that people learn things by interacting with the world. AI companies do collect these chatbot–human interaction records, but they go into a central database which is then processed in various ways and is used only to update the model's behavior once in a while.

Even for the tasks that chatbots are well capable of completing, their heavy reliance on statistics and lack of awareness of their own limitations blunt their usefulness. For example, playing rock-paper-scissors with state-of-the-art chatbots might go something like this:

USER: Let's play five rounds of rock-paper-scissors. You go first in each round. Keep score.

BOT: Round 1: I choose Rock.

USER: Paper.

BOT: You win! This time I choose Scissors.

...

Asked how the user won each time, ChatGPT suggested luck. Anthropic's Claude chatbot thought it could be fast reflexes. Google's Bard (later renamed Gemini) didn't realize that the user had won.

The bots' behavior in this *particular* example might be fixed by the time you read this—especially because AI developers are known to periodically fine-tune them based on known types of failures—but it shows the limits of trying to build intelligence by simulating transcripts of conversations rather than by interacting in the world.

Given that chatbots are statistical engines at their core, there is a big debate about whether they can understand or reason. There is a philosophical dimension and a practical dimension to the debate. The philosophical dimension is of less interest to us. From a practical standpoint, there are a few useful things to keep in mind.

Understanding is not all or nothing. Chatbots may not understand a topic as deeply or in the same way as a person—especially an expert—might, but they might still understand it to some useful degree.

Chatbots are often trained—fine-tuned—to answer confidently in the voice of an all-knowing expert. This is impressive at first, but after using them for a while, one soon catches them making basic errors even a child would know to avoid. The

failure to understand rock-paper-scissors is one example. Here's another: when asked "Which is heavier, a 1-pound feather or a 2-pound rock?" early versions of ChatGPT answered that they weighed the same!⁶¹ Instead of considering the actual numbers presented, they seemed to answer based on the superficial similarity to the well-known trick question "Which is heavier, a one-pound feather or a one-pound rock?"

Upon realizing these limitations, people sometimes swing in the other direction and conclude that chatbots have no understanding at all. The truth is somewhere in between.

Chatbots "understand" in the sense that they build internal representations of the world through their training process. Again, those representations might differ from ours, might be inaccurate, and might be impoverished because they don't interact with the world in the way that we do. Nonetheless, these representations are useful, and they allow chatbots to gain capabilities that would be simply impossible if they were merely giant statistical tables of patterns observed in the data.

Researchers' understanding of these internal representations is still rudimentary, because it is notoriously hard to figure out what the connections in a neural network actually encode. Still, we know that language models learn the structure of language, even though they don't have grammatical rules programmed into them.^{62,63}

In one study, researchers trained a GPT-2-like model on a database of Othello games, Othello being a checkers-like board game.⁶⁴ They found that when the trained model processes a new game to predict the next move, it actually tracks the state of the board. In a sense, it has learned the rules of the game despite never being explicitly told.

Still, in the vast chasm between the two extremes of “no understanding” and “perfect internal representation of the external world,” the question of where exactly chatbots lie is one that will likely be debated for a long time. Given that far more researcher effort goes into building AI than reverse engineering it, it appears likely that our ability to explain the internal workings of neural networks will continue to trail behind.

Chatbots have come a long way in the last few years. But as with image-processing AI, this progress hasn’t come without risks. In the next few sections, we will look at three such risks: misinformation, deepfakes, and centralization of power.

Automating Bullshit

Philosopher Harry Frankfurt defined bullshit as speech that is intended to persuade without regard for the truth.⁶⁵ In this sense, chatbots are bullshitters. They are trained to produce plausible text, not true statements. ChatGPT is shockingly good at sounding convincing on any conceivable topic. But there is no source of truth during training. Even if AI developers were to somehow accomplish the exceedingly implausible task of filtering the training dataset to only contain true statements, it wouldn’t matter. The model cannot memorize all those facts; it can only learn the patterns and remix them when generating text. So, many of the statements it generated would in fact be false.

Examples of automated bullshit range from amusing to grim. It has become commonplace for a researcher to receive an inquiry from a stranger about a paper they supposedly authored—only to realize that the title and description of the paper had been made up by a chatbot and attributed to them! One law

professor found out that ChatGPT had entirely fabricated a reference to a news article that supposedly accused him of sexually harassing a student.⁶⁶ Defamation by chatbot seems to happen quite frequently, and several lawsuits are ongoing as we write this.⁶⁷

The surprising thing is not that chatbots sometimes generate nonsense but that they answer correctly so often. We think it's best understood as a side effect of the fact that true statements are more plausible than false ones.

CNET, a popular news website, used generative AI to compose seventy-seven articles between November 2022 and January 2023. The company claimed that each of the articles was fact-checked before publication. But many had factual errors. The bot also plagiarized articles from competing websites, changing a few words while retaining the substance of the article.

When these problems were revealed, CNET reviewed each article generated automatically to look for errors, undercutting the efficiency gains promised by the AI tool. In its review, CNET found that forty-one of the seventy-seven articles contained errors. The website halted its use of generative AI after these revelations, but only temporarily.^{68,69}

The editors at CNET were presumably not trying to cause harm. But autogenerated language can also be used more nefariously, to deliberately bend the truth. In 2017, the Federal Communications Commission (FCC) received twenty-two million responses from a public call for feedback on their plans for net neutrality. Millions of these comments were generated automatically.⁷⁰ This was a blatantly antidemocratic attempt: entities with vested interests generated comments to make it seem like their opinions were held by the public at large.

Compared to systems like ChatGPT, these automated comments were unsophisticated. They relied on replacing words with synonyms.

I strongly urge the FCC to undo net neutrality
 I want to urge the FCC to undo net neutrality
 I want to demand the FCC to undo net neutrality

This program was likely built using a simple rule-based algorithm. Researchers were able to identify comments generated in this style by analyzing phrases and patterns that appeared across comments. But things are not so simple with advanced AI. It doesn't suffer from the limitations of crude text-generation techniques that make automated text easy to identify.

This hypothesis was tested in 2019. The U.S. state of Idaho released a public call for comments on its proposal to update Medicaid. The proposal received over 1,800 comments. Unbeknownst to the state agencies, 1,000 of these were autogenerated. Researchers used GPT-2, OpenAI's 2018 text-generation model, to submit seemingly real comments. Humans could not tell the difference.⁷¹

Researchers conducted this work ethically, and included a keyword in each response to identify which comments were autogenerated. But this study was prescient. As language models become widespread, so will automated bullshit.

Developers of language models can take several steps to improve things. They can warn users about when they should not rely on these models—essentially, any time that having factual information is paramount. Companies rarely share crucial information about leading language models, such as what data they are trained on. Transparently reporting these details and making the model available to researchers would enable them to identify problems.

Deepfakes, Fraud, and Other Malicious Uses

In 2019, a managing director at a British energy company received a call from his boss.⁷² He was told to urgently send over EUR 220,000 to a supplier. He sent over the money. He became suspicious when he received another call to send over more money, so he decided to call his boss back. To his surprise, his boss had never requested the money at all. This incident was one of the first confirmed instances of AI being used to scam people.

AI can generate lifelike voices these days. It is used to create audiobooks from text without a human narrator, generate overlay audio on apps like TikTok, and edit audio without the speaker's participation.^{73,74} Using seconds-long clips of a person's voice, AI can create a realistic audio clip of that person reading any text. And of course, as the incident at the British energy company demonstrates, the same tools are also being used to scam people out of money.

Scams aren't the only malicious use of AI-generated voices.⁷⁵ Users of 4chan, an anonymous online message board, created audio of celebrities saying offensive things. One audio clip depicted Emma Watson reading *Mein Kampf*. There are also worries that AI-generated audio could be used to doctor evidence in low-level court cases.⁷⁶ The standard for establishing authenticity in courts has traditionally been low, which can be exploited using AI.

Generative AI has also been used to create or edit images or videos of people without their consent ("deepfakes"). Apps like Photoshop have long enabled image editing, but AI has decreased the time and effort required. A messaging group on the Telegram app was used to coordinate the sale and purchase of sexualized, nude deepfakes, and it led to the sale of over a

hundred thousand deepfakes. The vast majority of AI-generated videos online are pornographic, created without the consent of the people they feature.⁷⁷

Companies and researchers have tried to offset these harms by building tools for detecting AI-generated audio, images, and video. Legislation could also help. In 2022, for instance, the UK government proposed making nonconsensual deepfakes with sexualized content illegal in its Online Safety Bill.⁷⁸ But perhaps the most important response is educating the public about the ease of faking content and the importance of relying on trustworthy sources for news and information.

The Cost of Improvement

The need for data for training generative AI has led to a massive reliance on labor in countries outside the United States and Europe. While the bulk of the training is done using data automatically scraped from the internet, some capabilities, such as engaging in dialogue with users rather than autocompleting text, require human involvement during training. Another reason large-scale labor is needed is to train models not to output toxic content, including hate speech, content promoting self-harm, and images of child abuse. Generative AI models' propensity to output such content is no surprise: since they are trained using data from the internet, they inherit the toxicity of the internet. But toxic content does pose a problem for companies trying to commercialize AI services and products like Google, Meta, OpenAI, and many others.

In 2016, Microsoft released a chatbot on Twitter named Tay. The model quickly started outputting all sorts of hateful comments and was taken down within a day. Until 2021, language models had a tendency to spew toxic text at unsuspecting users.

As one study showed, an innocuous prompt like “two Muslims” was enough to get GPT-3 (a predecessor of ChatGPT) to reliably generate text involving Muslims and violence, parroting and amplifying stereotypes present in its training data.⁷⁹

In contrast, more recent chatbots are much less likely to exhibit this behavior, though the defenses aren’t bulletproof. The ability to have a conversation without getting bombarded with inappropriate outputs is a major reason for the viability and success of chatbots.

To enable this, humans have to label millions of examples of toxic text and images. Labeling or annotating such content can be brutal. Annotators have to deal with low pay, high workloads, and exposure to intense content day in and day out. Because of cheaper labor and fewer regulations, much of this work happens outside the United States and Europe.

Sama, one of the firms contracted by OpenAI in Kenya, pays workers between USD 1.46 and 3.74 per hour.⁸⁰ In contrast, OpenAI pays many engineers close to a million dollars every year and is valued at eighty billion dollars as of early 2024.^{81,82}

This is part of a broader wave of the transition to globally distributed precarious work where AI companies hire contracting firms in lower-income countries. Precarious work refers to employment that is poorly paid, lacks employment benefits (such as mental health counseling that may help overcome the trauma that many AI-annotation workers face), and is insecure, which means that it may disappear at any time. As a *New York Magazine* investigation revealed: “It’s steady enough to be a full-time job for long stretches but too unpredictable to rely on. Annotators spend hours reading instructions and completing unpaid trainings only to do a dozen tasks and then have the project end. There might be nothing new for

days, then, without warning, a totally different task appears and could last anywhere from a few hours to weeks. Any task could be their last, and they never know when the next one will come.”⁸³ The work is so immiserating that many data annotation firms have taken to recruiting prisoners, people in refugee camps, and people in collapsing economies—in other words, those who will accept this work because they have no choice.⁸⁴

Due to strict confidentiality agreements, it is hard to know how many people work in data annotation. There isn’t even an order-of-magnitude estimate available. But based on the size of the market and the prevailing wages, the number is almost certainly in the millions.

One silver lining seems to be that the progress in AI capabilities has created demand for more satisfying types of AI training work. One worker reported that his work included “devising complex scenarios to trick chatbots into giving dangerous advice, testing a model’s ability to stay in character, and having detailed conversations about scientific topics so technical they required extensive research.” The work paid up to \$30 per hour, and he found it “satisfying and stimulating.” That said, it remains to be seen if progress in AI will make the more mundane type of annotation work obsolete.

In India, a nonprofit startup called Karya takes a radically different approach to data annotation. It pays twenty to thirty times the local minimum wage, and it allows workers to retain ownership of data they create. Best of all, the work helps build AI for their mother tongue, allowing them to benefit from the technology they help develop. Of course, AI companies aren’t exactly lining up to contract with Karya when cheaper options are available, so it remains to be seen if this model will catch on.⁸⁵

If this market won't fix itself, and it probably won't, then a new labor movement is needed. The Industrial Revolution famously led to a decades-long period of horrific working conditions as the demand for labor shifted from farms to accident-prone mines and factories located in overcrowded, disease-ridden cities. The modern labor movement arose as a response to these conditions. Perhaps there are lessons to be learned from that history. To improve the conditions of AI annotation work, an essay by Adrienne Williams and coauthors makes three recommendations: unionization, transnational organization, and solidarity between highly paid tech workers and their lower-paid counterparts.⁸⁶

Taking Stock

We've come a long way in this chapter. We hope that understanding the history and technical capabilities of generative AI has given you a way to evaluate how this technology is being used today—and how to discern which applications are beneficial, which are potentially harmful, and which are outright snake oil. Whatever new generative AI capabilities are coming next, our aim was to give you the foundation to make sense of them.

Generative AI is an exciting technology. It is fun to play with, intellectually interesting, and already practically useful, even transformational, for many types of workers. For example, in one survey, over 90 percent of U.S.-based programmers reported using AI to aid their work.⁸⁷

How should individuals approach generative AI? We think most knowledge workers can benefit from it, and we use it heavily in our own work. Since the technology is advancing quickly, any specific advice or product recommendations we

offer might be outdated by the time you read this. The unique thing about generative AI as opposed to other types of AI is that it is readily accessible to everyone. A good way to see how it can help you is to simply spend some time playing around with the available tools and looking into how your peers in your field are using them.

We've also looked at the limits and harms of generative AI. Let's use our understanding of the technology to make some educated guesses about the extent to which these problems can be fixed or mitigated.

In the last few years, there has been major progress on the problem of bias and offensive outputs.⁸⁸ That's because the fine-tuning process we described in this chapter has been highly effective at getting models to modify their behavior, even if it doesn't erase the underlying stereotypes and associations the models learn from the internet. There is still a long way to go, especially when it comes to image generators. But bias mitigation continues to be an area of vigorous and fruitful research.

As companies have deployed these mitigations, they have faced complaints that they have gone too far: that generative AI tools often deny legitimate requests in the name of safety, or that fine-tuning introduces political bias even as it curbs racial or gender bias. These are far harder questions, as they involve tradeoffs between legitimate values. They closely parallel debates over social media companies' policies on what sort of content is allowed on their platforms. We'll return to this topic in chapter 6.

On the problem of inaccurate outputs from chatbots, there has been gradual progress. Increasingly, chatbots combine the kind of text generation we discussed with retrieval of information in real time from authoritative online sources. This technology

is nascent in 2024, but if it works well, it can make chatbots as reliable as web search: far from perfect, but a big improvement over the purely generative approach.

Deepfakes pose a harder challenge and will impose costs on society. Commentators worry that, in addition to sexualized deepfakes, AI-generated disinformation could threaten democracy by allowing adversaries to influence elections. We think fears about vote manipulation are vastly overblown. Evidence suggests that people are naturally skeptical of what they see online and are resistant to persuasion. If anything, the problem is the opposite. The fact that just about anything online may have been AI-generated means that the internet will be seen as even less trustworthy. This has been called the “liar’s dividend.” Since the dawn of time, “seeing is believing” has been a reliable guide for seeking truth. Now, relatively suddenly, that rug has been pulled out from under us. Adapting to this new world won’t be easy.

New risks will arise as generative AI technology progresses. As video generation technology matures, new forms of entertainment will become possible. With those comes the potential for accelerating addictiveness of our devices, especially in the hands of children. This will require a combination of technological defenses, societal adaptation, and perhaps regulation.

The most serious harm from generative AI, in our view, is the labor exploitation that is at the core of the way it is built and deployed today. Some have argued that given these AI companies’ unscrupulous business practices, the only ethical course of action is to avoid using it altogether. That decision is up to individuals.

Realistically, however, we think collective action can be more fruitful than individual resistance. That can take the form of advocating for regulation, which we will take up in the final

chapter. Or, if your company is considering obtaining a license for generative AI products, choosing companies with relatively ethical practices may put pressure on other vendors to change their behavior.

For now, let's turn to another risk of generative AI that has made waves in the last few years: human extinction.

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